

wild type sequence

5' . . . GATTGCAGCTGCTGTTTTCCCACAATTCAGTATTGCCAGACACCGAGC . . .

primer for site-directed mutagenesis His103Val

5' GATTGCAGCTGCTGTTTTCGTGAATTCAGTATTGCCAGAC 3'

primer mixture for site-saturation mutagenesis His103Xxx

5' GCTGTTTTCNNKAATTCAGTATTGCCAGAC 3'